

QueleaGuard

Building the intelligence foundation for adaptive Quelea crop protection

From a family rice farm to an AI research and data-engineering journey.



I watched my family's rice farm disappear.

Kisumu, Kenya

Rice farming for
approximately 16 years

2024 — The Breaking Point

Five people guarding fields, 6
AM to 6 PM, every day

Potential Harvest

50+ bags

Actual Harvest

12 bags — premature

My family eventually stopped
farming rice.



A farmer cannot fight a persistent bird threat with five people.



Continuous Monitoring

Large field areas demand coverage no human team can sustain



Unsustainable Labour

Manual deterrence is intensive, costly, and ultimately ineffective



Slow Reaction Time

Flock movement outpaces any human response capability



Limited Intelligence

No actionable information about bird behaviour or movement patterns

How can we continuously detect, understand and respond to Quelea activity at a scale humans cannot?

My first answer was Agri-Shield.

LONG-TERM VISION

1

DETECT

AI watchtowers + IoT field units + environmental sensors

2

RESPOND

Drones + automated deterrence + central intelligence

3

LEARN

Machine learning feeds back into detection and response



But I was trying to build the top of the system first.



1

2

3

1 Data

The irreducible foundation

2 Models

Models require data

3 Intelligence

Intelligence requires models

An intelligent system needs intelligence. So I went back to the beginning.

QueleaGuard became the foundation.

QueleaGuard — Now

Data + research + intelligence foundation

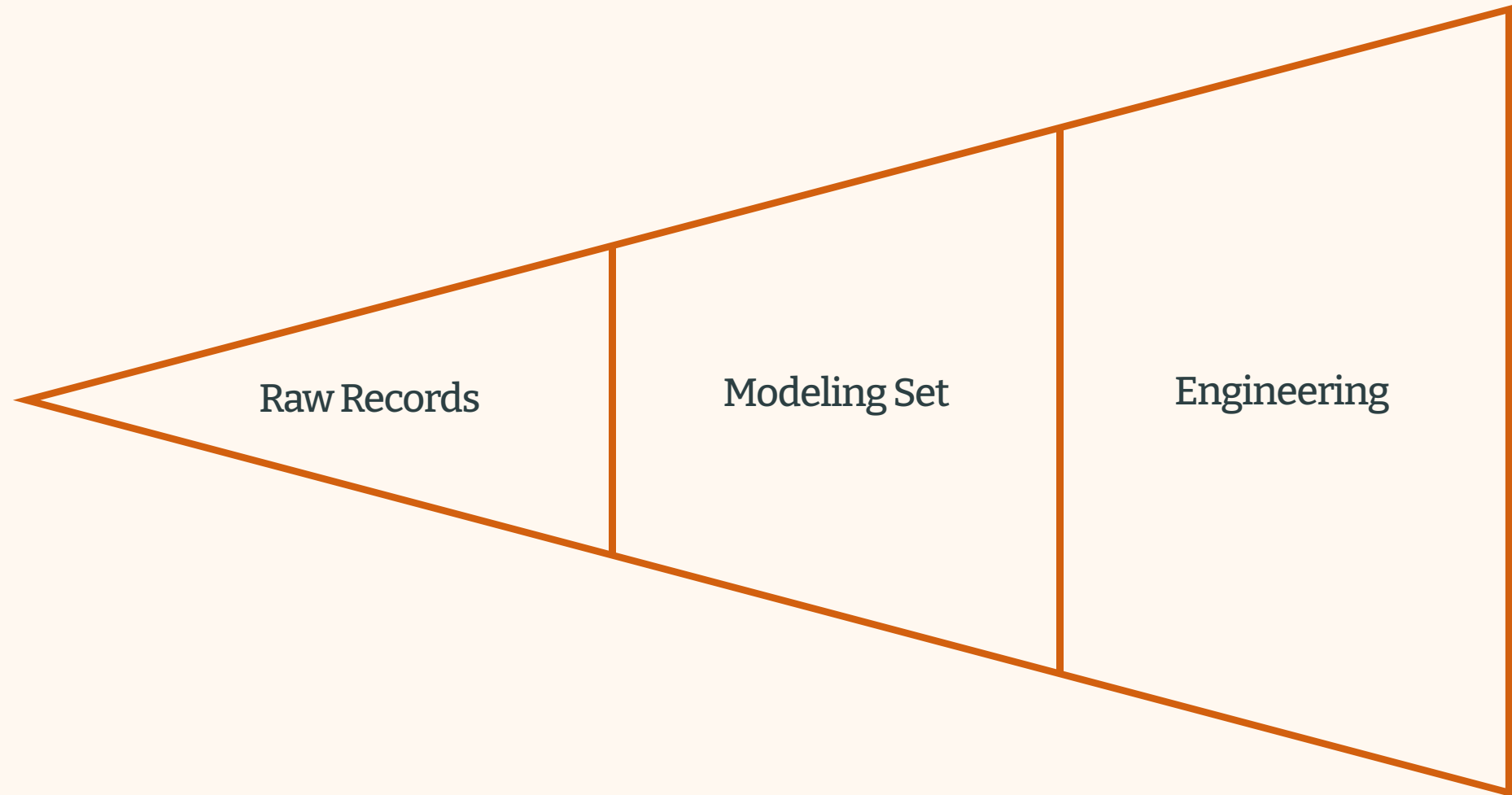
Understanding Quelea occurrence, spatial context, environmental patterns, and data quality.

Agri-Shield — Long-Term Vision

The autonomous crop-protection system that QueleaGuard makes possible.

Every model, every sensor, every intervention — built on the foundation we are engineering today.

The first problem wasn't machine learning. It was **data**.



Quelea occurrence records, rainfall, vegetation, temperature, terrain, hydrology, spatial boundaries, and temporal information — all fragmented, all requiring careful engineering before a single model could be trained.

Five weeks of turning fragmented information into a research dataset.

01

Occurrence Data → Spatial
Framework → Grid

02

Presence / Pseudo-absence
Extraction

03

Environmental Data Alignment

04

Temporal Synchronization

05

Quality Control → Modelling Dataset



Now I'm going to show you exactly how we built this — live, in the repository and notebook.

Building the dataset was harder than building the model.

Data Scarcity

Limited direct *Quelea* observations available for Kenya

Spatial Scale

Choosing an analytical framework that reflects ecological reality

Temporal Alignment

Matching observations precisely with environmental conditions

Pseudo-absences

No conventional absence data — methodology required careful justification

Remote Sensing

Quality and coverage issues across environmental datasets

Data Quality

Invalid values detected, investigated, and corrected

The dataset became a research problem of its own.

Data availability determines what can responsibly be modelled

Spatial and temporal alignment matter as much as model selection

Presence-only observations require careful methodology

Environmental datasets are not automatically trustworthy

We chose research integrity over demo theatre.

What comes next?

